Tarun Gupta

B.Tech., Computer Science and Engineering
Indian Institute Of Technology (IIT), Indore

tarungupta360@gmail.com
tarun360.github.io (personal website)
linkedin.com/in/tarun-gupta-116860178

Education

B.Tech., Computer Science and Engineering **2018-2022**
Indian Institute of Technology (IIT), Indore **CGPA: 9.81/10**
Department rank 1, Institute rank 2

Experience

Research associate at IISc *July 2024 - Present*
LLMs, NLP Paper

- Conducted research on plagiarism in AI-generated research, identifying that 24% of LLM-generated documents contain unattributed content, and demonstrated through controlled experiments that automated plagiarism detection tools are inadequate at catching deliberately plagiarized ideas.
- Researched the root causes of position bias and lost-in-the-middle problem in LLMs, investigating factors contributing to these performance limitations.
- Researched philosophical arguments for and against the "world-models" research-direction in AI.

Senior Member Technical at D. E. Shaw (Hedge Fund) *July 2022 - June 2024*
Software Development, High-Performance Computing

- Worked on developing an internal High-Performance Cluster (HPC) scheduling software.
- Served as primary full-stack developer for LLM-powered employee review automation software.
- Contributed to the open-source container technology tool Podman by adding a feature to encrypt container-images (GitHub PR link).
- Awarded **super-star promotion** to Senior Member Technical (L2) in 1.5 years.

Software Development Intern at D. E. Shaw (Hedge Fund) *May 2021 - July 2021*
Software development, React, Kafka, Java

- Designed and implemented a graph visualization tool using GoJS to enhance traders' workflow efficiency with an intuitive representation of their complex workflows.

Research Intern at NTU Singapore and Skit.ai *Dec 2021 - June 2022*
Speech Processing, Transformers Paper 1, Paper 2

- Built a bi-encoder transformer mixture model to estimate speaker age and height from speech signal, leading to a relative improvement of 18.5% in male age estimation and 8.6% in female age estimation over the state-of-the-art benchmarks on the TIMIT dataset (refer to Paper 1).
- Developed a new data augmentation method for spoken language identification, Map-Mix, that leverages model training dynamics of individual data points to improve sampling for latent mixup. Map-Mix improves weighted F1 scores by 2% compared to the random mixup baseline (refer to Paper 2).

Research Intern at Carnegie Mellon University *Jan 2021 - May 2021*
Computer vision, Self-supervised learning Paper

- Built a contrastive self-supervised learning (CSSL) pipeline for macromolecular structure classification using cryo-electron tomography (cryo-ET) data.
- Extended techniques such as SimCLR, MoCo, and SwAV to the cryo-ET domain, achieving a significant improvement over non-self-supervised learning methods.

Research Assistant at IIT Indore *Jan 2020 - May 2020*
SVM, Convex programming Paper

- Designed a novel plane-based clustering algorithm based on the principles of SVM, utilizing pinball loss to improve generalization performance on noise-corrupted datasets. Used Concave-Convex Procedure (CCCP) for optimization.
- The proposed algorithm improved clustering accuracy by $\sim$1% on noise-corrupted UCI datasets compared to existing plane-based clustering algorithms.

Publications

Gupta, T. and Pruthi, D., 2025. Beyond World Models: Rethinking Understanding in AI models. In submission to EMNLP 2025.

Gupta, T. and Pruthi, D., 2025. All That Glitters is Not Novel: Plagiarism in AI Generated Research. ACL 2025.

Rajaa, S., Anandan, K., Dalmia, S., **Gupta, T**. and Chng, E.S., 2023, June. Improving Spoken Language Identification with Map-Mix. International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2023.

Gupta, T., Truong, T.D., Anh, T.T., Chng, E.S. (2022) Estimation of speaker age and height from speech signal using bi-encoder transformer mixture model. Interspeech 2022.

Gupta, T., He, X., Uddin, M.R., Zeng, X., Zhou, A., Zhang, J., Freyberg, Z. and Xu, M., 2022. Self-supervised learning for macromolecular structure classification based on cryo-electron tomograms. Frontiers in Physiology.

Tanveer, M., **Gupta, T.**, Shah, M. and Richhariya, B., 2021. Sparse twin support vector clustering using pinball loss. IEEE Journal of Biomedical and Health Informatics (JBHI).

Achievements

- **CSE Department Rank 1** and **Institute Rank 2** at IIT Indore (based on CGPA).
- **Best Undergraduate Researcher Award, 2021**.
- **Awarded AP grades**: for exceptional performance in **5** courses.
- **Inter IIT Tech Meet 9.0 (2021)**: Awarded **Silver medal** in Bridgei2i's NLP competition.
- **Inter IIT Tech Meet 8.0 (2020)**: Awarded **Bronze medal** in BITGRIT's data-science competition.
- **JEE Advanced 2018**: Secured All India Rank 1055 among 150,000 candidates (top 0.7%).

Key Projects

Satellite video prediction (in collaboration with REint.co) *August 2023 - September 2023*

Computer Vision, Spatio-Temporal Predictive Learning

- Implemented and tested various spatio-temporal prediction models, such as ConvLSTM and SimVP, on real-world satellite data. The predicted frames can be fed to models such as MetNet-2 for weather prediction.
- Explored and employed PyTorch DDP for multi-node and multi-GPU training.

Parallelizing Red Deer Algorithm (RDA) – A Nature Inspired Meta-heuristic Algorithm *April 2021*

Parallel Programming, MPI GitHub Link

- Implemented Red Deer Algorithm (RDA) for solving the Traveling Salesman Problem (TSP). RDA is a meta-heuristic algorithm inspired by the unique mating process of Scottish red deer.
- Used Message Passing Interface (MPI) for parallelizing the RDA algorithm, achieving a speed-up factor up to 4.

Bridgei2i's NLP competition at Inter IIT Tech Meet 9.0 (2021) *March 2021*

Awarded Silver Medal GitHub Link

- *Theme identification task*: Applied DistilBERT, a distilled version of the BERT model, for fast and scalable binary classification of tweets and articles, achieving an F1 score of 0.89.
- *Sentiment analysis task*: Utilized Ada-BERT transformer model fine-tuned for aspect-based sentiment analysis to identify the sentiments associated with a particular brand.
- *Headline generation task*: Utilized Pegasus, T5, and BART models, achieving 37% average similarity score.

Adversarial Attacks on Brain Tumor Segmentation Models *April 2020 - June 2020*

AI for Medicine GitHub Link

- Implemented a 3D-UNet model using the TensorFlow library and trained it on BraTS brain MRI data for brain tumor segmentation.
- Tested the model's robustness by implementing adversarial attacks such as Fast-Gradient Sign Method (FGSM), Iterative Fast-Gradient Sign Method (iFGSM), and Carlini & Wagner (CW) attack.

Cache-Oblivious Algorithms *May 2020 - July 2020*

Design and Analysis of Algorithms GitHub Link

- Implemented various cache-oblivious algorithms such as Van-Emde-Boas search tree, Funnel-sort, and Median of medians algorithm.
- Analyzed the memory-transfer complexity of the above algorithms. Used Valgrind to calculate the cache hit-miss ratio of cache oblivious algorithms and compared it with their cache ignorant counterparts.

Relevant coursework

- **Computer Science**: Computer Vision, Cryptography & Network Security, Advanced Algorithms, Machine Learning, Compiler Techniques, Computational Intelligence, Computer Networks, Computer Graphics & Visualization, Optimization Algorithms & Techniques, Operating Systems, Computer Architecture, Parallel Computing, Design & Analysis of Algorithms, Software Engineering, Automata Theory & Logic, Logic Design, Data Structures & Algorithms, Database & Information Systems†, Discrete Mathematical Structures.
- **Mathematics**: Numerical Methods†, Complex Analysis & Differential Equations II, Linear Algebra & Differential Equations I, Calculus.
- **Other**: Psychology†, Physics†, Chemistry†.
- **AI for Medicine (Coursera)**: AI For Medical Diagnosis, AI For Medical Prognosis & AI For Medical Treatment.
- **Quantum Computation (David Deutsch)**.

(† indicates an exceptional grade (AP) achieved in the course.)

Technical Skills

- **Programming**: Python, C++, C, POSIX C, Go, Java, SQL, VerilogHDL, HTML, CSS.
- **Tools/Libraries**: PyTorch, TensorFlow, Container technology (Podman, Docker), LaTeX.

Positions of Responsibility / Extra Curricular Activities

- **Head, Cynaptics Club (AI and ML student club of IIT Indore)**: Led, mentored, and taught club members and students. Organized competitions and administered tutorials. *Aug 2020 - Jun 2021*
- **Department Undergraduate Representative**, Department of Mathematics. *Feb 2020 - Oct 2020*